UNIT 1 · STANDARD 6.NS.2

Divide Multi-Digit Numbers

Lesson 1-4

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In $156 \div 12 = 13$, the dividend is 156 — it is the total being split up

Dividend

The number you are splitting up in a division problem.

In $156 \div 12 = 13$, the divisor is 12 — it is the size of each group

Divisor

The number you split by in a division problem.

In $156 \div 12 = 13$, the quotient is 13 — there are 13 groups

Quotient

The answer when you divide.

$17 \div 5 = 3$ R 2 means 3 groups of 5 with 2 left over

Remainder

What is left over when a number does not divide evenly.

$1{,}344 \div 12$: first $12 \times 100 = 1{,}200$, then $12 \times 12 = 144$. Quotient $= 100 + 12 = 112$

Partial quotients

A way to divide by breaking the problem into smaller, easier steps.

Key Ideas & Notes

- The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation.
- How many bars does each section receive?
- Break apart the division 1,344 ÷ 12 using partial quotients. Sort each partial product — does it help build toward the answer, or is it NOT a step in this division?

Think About It

- What total amount needs to be divided?
- How many groups are the supplies being split into?
- Will each section get the same amount with none left over?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The station must divide 1,344 nutrition bars equally among 12 crew sections. Before dividing, about how many bars should each section get, and how do you know?

Level 1 support

Start here: Estimating with friendly numbers checks whether the quotient is reasonable.

Each section gets about ____ bars because 1,344 is close to ____.

Cada sección recibe unas ____ barras porque 1,344 está cerca de ____.

I estimated by ____.

Estimé al ____.

Word bank: dividend | divisor | quotient | estimate | partial quotients

Listen for: *Students estimate using 1,200 ÷ 12 = 100 (or similar) to predict roughly 100+ bars per section, identifying 1,344 as the dividend and 12 as the divisor.*

Level 2

Will 1,344 divide evenly by 12, or will there be a remainder? Predict before computing and justify your reasoning.

- I predict the remainder is ____ because ____.
- 1,344 is divisible by 12 since ____.

EXPLORE

You used partial quotients for 1,344 ÷ 12. What steps did you use, and how do they build to the final quotient?

Level 1 support

Start here: Partial quotients break a hard division into easy chunks you add together.

First I took 12 × ____ = 1,200, then 12 × ____ = 144.

Primero usé 12 × ____ = 1,200, luego 12 × ____ = 144.

The quotient is ____ because I added the partial quotients ____.

El cociente es ____ porque sumé los cocientes parciales ____.

Word bank: **partial quotients** **dividend** **divisor** **quotient** **remainder**

Listen for: *Students explain 12 × 100 = 1,200, then 144 left, 12 × 12 = 144, so the quotient is 100 + 12 = 112 with no remainder.*

Level 2

Why does it not matter whether you start with 12 × 100 or 12 × 50 in partial quotients? Will you reach the same quotient?

- It does not matter where I start because ____.
- Different first chunks still give 112 since ____.

CONNECT

A school divides 1,680 pencils equally among 24 classrooms. How does multi-digit division tell each classroom's share, and how can you check it?

Level 1 support

Start here: Dividing 1,680 by 24 gives each classroom's count; multiply back to check.

Each classroom gets ____ pencils because 1,680 ÷ 24 = ____.

Cada salón recibe ____ lápices porque 1,680 ÷ 24 = ____.

I can check by multiplying ____ × ____ = 1,680.

Puedo comprobar multiplicando ____ × ____ = 1,680.

Word bank: **dividend** **divisor** **quotient** **partial quotients** **equally**

Listen for: *Students compute 1,680 ÷ 24 = 70 and verify with 24 × 70 = 1,680, explaining the quotient is the per-classroom amount.*

Level 2

If the school had 1,700 pencils instead, what would the quotient and remainder be, and what would the remainder mean in this situation?

- With 1,700 pencils the quotient is ____ and the remainder is ____.
- The remainder means ____ in real life because ____.

REFLECT

In a division problem, what is the difference between the quotient and the remainder, and what does each one tell you?

Level 1 support

Start here: The quotient is the size of each equal share; the remainder is what is left over.

The quotient tells me ___, and the remainder tells me ___.

El cociente me dice ___, y el residuo me dice ___.

A remainder of 0 means ___.

Un residuo de 0 significa ___.

Word bank: **quotient** **remainder** **dividend** **divisor** **divide**

Listen for: *Students explain the quotient is the number in each group while the remainder is the amount left that cannot be shared equally, and a remainder of 0 means it divides evenly.*

Level 2

When is it correct to round a quotient UP because of a remainder, and when should you drop the remainder? Give an example of each.

- I round up when ___, for example ___.
- I drop the remainder when ___, for example ___.

Guided Examples

Example 1

What is 936 ÷ 12?

Solution: 12 × 78 = 936. You can verify: 12 × 70 = 840, 12 × 8 = 96, 840 + 96 = 936.

Answer: A. 78

Example 2

What is 2,485 ÷ 5?

Solution: 5 × 497 = 2,485. Check: 5 × 400 = 2,000, 5 × 90 = 450, 5 × 7 = 35. 2,000 + 450 + 35 = 2,485.

Answer: A. 497

Example 3

What is 756 ÷ 9?

Solution: 9 × 84 = 756. Check: 9 × 80 = 720, 9 × 4 = 36, 720 + 36 = 756.

Answer: A. 84

Write About the Math

The Writing Revolution

I can explain my division using the words dividend, divisor, quotient, and remainder.

1. Kernel Sentence

subject + verb

Model: Partial quotients is a way to divide by breaking the problem into smaller, easier steps.
Cocientes parciales es una manera de dividir separando el problema en pasos más fáciles.

Write a kernel sentence about partial quotients. Use a subject and a verb.
Escribe una oración base sobre cocientes parciales. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Partial quotients matters in math
Cocientes parciales importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because / ***porque***

Partial quotients matters in math because ____.
Cocientes parciales importa en matemáticas porque ____.

____

but / ***pero***

Partial quotients matters in math, but ____.
Cocientes parciales importa en matemáticas, pero ____.

____

so / ***entonces***

Partial quotients matters in math, so ____.
Cocientes parciales importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about partial quotients.
Di un hecho verdadero sobre partial quotients.

Partial quotients ____.

Question
Pregunta

Ask a question about partial quotients.
Haz una pregunta sobre partial quotients.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about partial quotients.
Muestra entusiasmo sobre partial quotients.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with partial quotients.
Dile a un compañero qué hacer con partial quotients.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

Each section gets about ____ bars because 1,344 is close to ____.
Cada sección recibe unas ____ barras porque 1,344 está cerca de ____.

I estimated by ____.
Estimé al ____.

First I took 12 × ____ = 1,200, then 12 × ____ = 144.
Primero usé 12 × ____ = 1,200, luego 12 × ____ = 144.

Try It

Solve on your own. Check the answer key when you are done.

1. What is 1,125 ÷ 9?

A. 125

B. 115

C. 135

D. 124

Show your work:

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2. A warehouse has 2,184 items to pack into boxes of 14. How many boxes are needed?

A. 156 boxes

B. 146 boxes

C. 166 boxes

D. 155 boxes

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A school has 1,680 pencils to distribute equally to 24 classrooms. Show how to solve this using partial quotients. Explain each step and check your answer by multiplying.

Sentence starter: 1,680 ÷ 24 = ____. First I used 24 × ____ = ____, then 24 × ____ = ____. The partial quotients add to ____. I checked by multiplying: 24 × ____ = 1,680.

Show your work:

Reflect — Exit Ticket

What is 2,352 ÷ 16?

A. 147

B. 137

C. 157

D. 146

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 125 — *9 × 125 = 1,125. Check: 9 × 100 = 900, 9 × 25 = 225, 900 + 225 = 1,125.*
2. **Try It 2:** A. 156 boxes — *2,184 ÷ 14 = 156. Check: 14 × 156 = 14 × 150 + 14 × 6 = 2,100 + 84 = 2,184.*
3. **Exit Ticket:** A. 147 — *16 × 147 = 2,352. Check: 16 × 100 = 1,600, 16 × 40 = 640, 16 × 7 = 112. 1,600 + 640 + 112 = 2,352.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Partial quotients is a way to divide by breaking the problem into smaller, easier steps.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).